Applied Statistics · Module 4 · PAGE 1 — OVERVIEW

Simple linear regression models the relationship between a predictor X and an outcome Y. The equation is Y_hat = beta_0 + beta_1 * X. The intercept beta_0 is the predicted Y when X is zero; the slope beta_1 is the expected change in Y for a one-unit increase in X. A prediction is an estimate, not a guarantee.

Applied Statistics · Module 4 · PAGE 2 — FIT AND RESIDUALS

A residual equals observed Y minus predicted Y_hat. Ordinary least squares fits beta_0 and beta_1 by minimizing the sum of squared residuals. Large residuals suggest poor fit for a particular observation. Inspecting a residual plot helps identify nonlinear patterns or changing error variance.

Applied Statistics · Module 4 · PAGE 3 — ASSUMPTIONS AND PITFALL

A basic linear regression analysis assumes a linear mean relationship and makes assumptions about residual behavior. Correlation does not prove causation. Extrapolating far outside the observed X range is risky. Check data quality, outliers and whether an omitted variable might explain the apparent relationship.

Applied Statistics · Module 4 · PAGE 4 — PRACTICE TASK

A dataset records hours studied (X) and quiz score (Y). Write the model equation, explain slope versus intercept in plain language, sketch what a residual means, and describe two ways the model could be misleading. A suggested study session sequence is: review concept (25 min), work examples (35 min), check assumptions (20 min), complete assignment draft (70 min).